



United International University
School of Science and Engineering
Department of Computer Science and Engineering
Mid Term Examination || Trimester: Spring – 2026
Course: Math -2205 (Probability & Statistics)
Total marks – 30 || Duration – 1 hour 30 minutes

[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

- Q1** (a) A company is comparing the performance consistency of two sales teams based on their monthly sales (in thousand BDT) over a certain period.
Team A: Mean sales = 50, Standard deviation = 5
Team B: Mean sales = 60, Standard deviation = 12
- (i) Calculate the **coefficient of variation (CV)** for both Team A and Team B. [2]
(ii) From a managerial perspective, discuss **which team is more reliable** and why? [2]

- Q2** (a) A researcher is analyzing the **distribution of weekly online study hours** of students in a university. The following summarized statistics (about the mean) are obtained:
- Mean (μ) = 10 hours
 - Second central moment (μ_2) = 16
 - Third central moment (μ_3) = 8
 - Fourth central moment (μ_4) = 768
- (i) Calculate the **coefficient of skewness** using moment-based formula and interpret the result. [2]
(ii) Calculate the **kurtosis** and comment on the **nature of the distribution** (leptokurtic, mesokurtic, or platykurtic). [2]
- (b) A teacher recorded the test scores (out of 100) of 9 students: 52, 50, 52, 45, 58, 60, 82, 65, 98
Find the five-number summary and Identify if there is any **outlier** using the IQR method. [4]

- Q3** (a) A researcher is studying the relationship between temperature ($^{\circ}\text{C}$) and electricity consumption (in units) in a household over 6 days. The data are given below:

Days	Temperature (x)	Electricity Consumption (y)
1	20	28
2	23	31
3	24	37
4	27	36
5	28	40
6	31	41

- (i) Calculate the **coefficient of correlation (r)** between temperature and electricity consumption and interpret the result. [4]
(ii) Find the **regression line of y on x** in the form: $y = a + bx$ and interpret the constant **a** and **b**. [4]
(iii) Using the regression equation, estimate the electricity consumption when the temperature is 25°C [1]

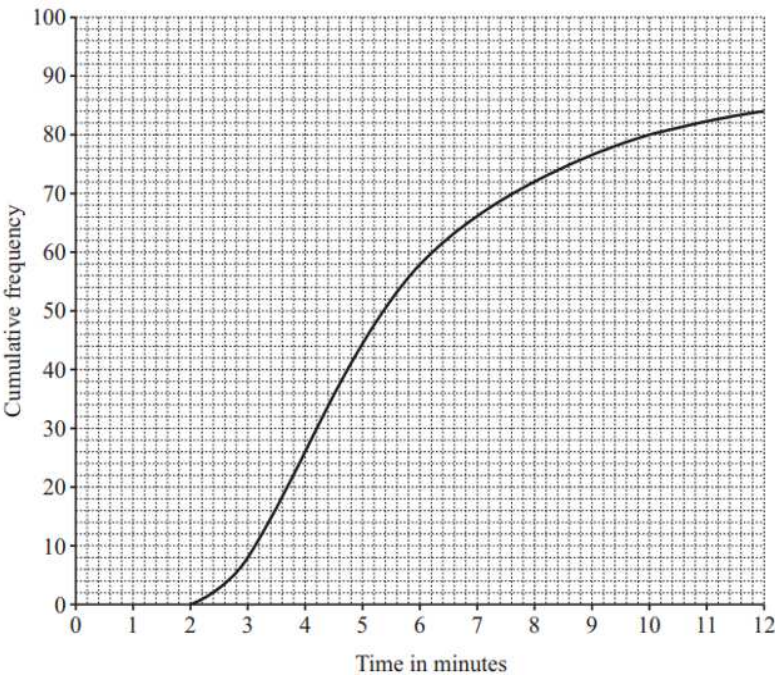
Q4 (a) The back-to-back stem-and-leaf diagram shows the annual salaries in dollars of 27 employees at each of the two companies, **Browns and Greens**.

Browns									Greens										
(3)	9 8 4							30	6 8	(2)									
(7)	8	8	5	3	3	1	0	31	2	4	5	8	(4)						
(7)	9	7	6	4	2	2	0	32	3	6	6	7	9	(5)					
(6)	8 7 5				5	3	1	33	1	1	3	7	8	8	9	9	(8)		
(3)					4	2	2	34	0	2	3	5	6	9	(6)				
(1)								7	35	3	7	(2)							

Key: 6|32|7 means \$32 600 for Browns and \$32 700 for Greens.

- (i) Find the **median and interquartile** range for the annual salaries of employees at Browns.
 [3]
- (ii) Which group shows **greater variability**? Justify.
 [2]

(b) 84 people attempt a particular puzzle. The times taken, in minutes, to complete the puzzle are recorded. These times are represented in the cumulative frequency graph below.



- () Use the graph to estimate how many people took between 4 to 8 minutes to complete the puzzle.
 [2]
- (b) 25% of the people took more than standard time. How many people took more than standard time.
 [2]

Formulae:	For the regression line: $y = a + bx$	$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$ and $b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$
	Coefficient of correlation	$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$
	Spearman's rank correlation coefficient:	$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$